Jared Duong

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Education

University of Guelph **Sep 2024 – Apr 2029**
Bachelor of Engineering in Computer Engineering (Co-op) *Guelph, ON*

Technical Skills

Languages: Python, Java, C, C++, C#, SQL, JavaScript, HTML/CSS, VHDL
Frameworks & Libraries: pandas, openpyxl, NumPy, JavaFX, JUnit, .NET, Unity
Software & Tools: REST APIs, MySQL, Git, GitHub, VS Code, Visual Studio, IntelliJ IDEA, ModelSim, Vivado, Microsoft Excel

Experience

Software Integration Developer **Jan 2026 – Mar 2026**
Appleby North Paint Inc. *Remote*

- Developed a Python data-processing pipeline using pandas and openpyxl to process Shopify and Decor Fusion exports, generating product and inventory feeds for **206 SKUs**.
- Implemented SKU-based matching across three Excel datasets, linking **154 products** to pricing records and identifying **52 unmatched SKUs**.
- Automated inventory availability classification and generated diagnostic reports flagging **146 pricing discrepancies** for review.

Summer Production Student **May 2025 – Jun 2025**
Hartmann Canada Inc. *Brantford, ON*

- Operated and monitored manufacturing equipment, checking panel indicators and gauges to identify potential production issues.
- Performed quality-control inspections to verify product weight, colour, and functionality against established production specifications.
- Recorded inspection observations, communicated production concerns to team members, and followed workplace safety procedures.

Projects

University Management System | *Java, JavaFX, MySQL, SQL* **Mar 2025 – Apr 2025**

- Developed a JavaFX application backed by MySQL to manage student records and course information, designed to support **20+ users**.
- Designed a normalized relational database with primary and foreign keys, implementing SQL queries for data retrieval, filtering, and updates.
- Applied object-oriented design to separate application logic from database operations, testing and debugging data-management workflows.

MIPS-Based 16-Bit CPU System | *VHDL, ModelSim, Vivado* **Jan 2026 – Mar 2026**

- Co-developed a **16-bit single-cycle CPU** in VHDL with an ALU, **16-register** file, hardwired control unit, and separate instruction and data memories.
- Implemented **10 instructions** for arithmetic, logic, memory access, and control flow, including branching and jumping operations.
- Verified instruction execution using ModelSim testbenches and waveform analysis, debugging write-back paths and program-counter logic.

4x4x4 LED Cube with Custom PCB | *Arduino, C/C++, PCB Design* **Mar 2025 – Apr 2025**

- Designed and soldered a **64-LED cube** using 74HC04 and SN74HC08 logic ICs, a custom PCB, and a 3D-printed base.
- Programmed LED patterns on an Arduino Nano in C/C++ using multiplexing, layer-scanning, and timing logic.

Engineering Involvement

Mechanical Team Member **Sep 2024 – Present**
University of Guelph Robotics Team *Guelph, ON*

- Currently designing a camera mount in SOLIDWORKS for the competition rover, focusing on component fit and mechanical integration.
- Assisted with rover drivetrain and wheel-subsystem assembly and collaborated with electrical and software subteams on integration, design reviews, and testing.